Statistics

Kesondra Key

2026-03-29

IMPORTANT: Change "E:/Github/diurnal-divergence-factor/data/" to your own file location in the setup and on line 263

Purpose: Run statistics and generate tables Note: Not all of the generated .csv's were used!

General information about the dataframe like number of sites and number of droughts per category

```
## # A tibble: 2 x 2
##   drought_cat n_periods
##   <chr>           <int>
## 1 LS                431
## 2 MS                 67
```

```
## # A tibble: 2 x 4
##   drought_cat n_sites median_periods_per_site max_periods_per_site
##   <chr>         <int>                   <dbl>                <int>
## 1 LS               40                     9.5                   24
## 2 MS               15                     5                     10
```

```
## [1] 40
```

```
## [1] 49644
```

```
##    sitename              Date                 Year         Latitude
##  Length:49644       Min.   :2001-05-01   Min.   :2001   Min.   :27.38
##  Class :character   1st Qu.:2010-07-11   1st Qu.:2010   1st Qu.:31.79
##  Mode  :character   Median :2015-06-18   Median :2015   Median :35.97
##                     Mean   :2014-11-06   Mean   :2014   Mean   :36.57
##                     3rd Qu.:2019-08-04   3rd Qu.:2019   3rd Qu.:39.10
##                     Max.   :2024-10-31   Max.   :2024   Max.   :45.82
##
##    Longitude           Veg               mean_AI          AI_cat
##  Min.   :-121.95   Length:49644       Min.   :0.1461   Length:49644
##  1st Qu.:-110.87   Class :character   1st Qu.:0.2354   Class :character
##  Median :-109.39   Mode  :character   Median :0.5392   Mode  :character
##  Mean   :-101.71                      Mean   :0.6501
##  3rd Qu.: -85.31                      3rd Qu.:0.9573
##  Max.   : -76.56                      Max.   :2.2119
##
##    Terrain          data_source              ET                  VPD
##  Length:49644       Length:49644       Min.   :-4.381e-05   Min.   :0.3142
```

```
##  Class :character   Class :character   1st Qu.: 1.471e-05   1st Qu.:1.2126
##  Mode  :character   Mode  :character   Median : 4.329e-05   Median :1.7479
##                                        Mean   : 4.844e-05   Mean   :2.0286
##                                        3rd Qu.: 7.654e-05   3rd Qu.:2.7002
##                                        Max.   : 2.443e-04   Max.   :6.6664
##
##       DDF             Drought          drought_cat          Cluster
##  Min.   :-11.000   Length:49644       Length:49644       Min.   : 1.0
##  1st Qu.: -4.500   Class :character   Class :character   1st Qu.: 6.0
##  Median : -3.000   Mode  :character   Mode  :character   Median :12.0
##  Mean   : -3.122                                         Mean   :13.8
##  3rd Qu.: -1.500                                         3rd Qu.:19.0
##  Max.   :  7.000                                         Max.   :45.0
##                                                          NA's   :38687
##    DroughtNo       ClusterStart        min_scaled_SWC
##  Min.   : 1.000   Min.   :2001-07-11   Min.   :-3.301
##  1st Qu.: 5.000   1st Qu.:2010-07-23   1st Qu.:-1.533
##  Median : 9.000   Median :2014-05-20   Median :-1.245
##  Mean   : 9.889   Mean   :2014-10-01   Mean   :-1.336
##  3rd Qu.:14.000   3rd Qu.:2019-09-07   3rd Qu.:-1.060
##  Max.   :28.000   Max.   :2024-10-13   Max.   :-0.818
##  NA's   :38687    NA's   :38687        NA's   :38687
```

```
## [1] 415
```

```
## [1] 498
```

```
## 40 US flux tower sites were selected, covering 498 unique drought periods across 415 site-years.
```

```
## # A tibble: 3 x 2
##   drought_cat n_sites
##   <chr>         <int>
## 1 LS               40
## 2 ND               40
## 3 MS               15
```

```
## # A tibble: 2 x 2
##   drought_cat n_periods
##   <chr>           <int>
## 1 LS                431
## 2 MS                 67
```

```
## # A tibble: 2 x 2
##   drought_cat n_periods
##   <chr>           <int>
## 1 LS                431
## 2 MS                 67
```

```
## # A tibble: 2 x 2
##   AI_cat n_sites
##   <chr>    <int>
## 1 Humid       27
## 2 Arid        13
```

```
## # A tibble: 4 x 2
##   Veg   n_sites
##   <chr>   <int>
## 1 GRA        14
## 2 SHR        11
## 3 ENF         8
## 4 DBF         7
```

```
## # A tibble: 6 x 3
##   AI_cat drought_cat n_sites
##   <chr>  <chr>         <int>
## 1 Arid   LS               13
## 2 Arid   MS                1
## 3 Arid   ND               13
## 4 Humid  LS               27
## 5 Humid  MS               14
## 6 Humid  ND               27
```

```
## # A tibble: 12 x 3
##    Veg   drought_cat n_sites
##    <chr> <chr>         <int>
##  1 DBF   LS                7
##  2 DBF   MS                6
##  3 DBF   ND                7
##  4 ENF   LS                8
##  5 ENF   MS                2
##  6 ENF   ND                8
##  7 GRA   LS               14
##  8 GRA   MS                4
##  9 GRA   ND               14
## 10 SHR   LS               11
## 11 SHR   MS                3
## 12 SHR   ND               11
```

```
## # A tibble: 40 x 8
##    sitename Latitude Longitude Veg   mean_AI AI_cat Terrain          data_source
##    <chr>       <dbl>     <dbl> <chr>   <dbl> <chr>  <chr>            <chr>
##  1 US-AUD       31.6    -111.  GRA     0.235 Arid   <NA>             Ameriflux
##  2 US-CHR       35.9     -84.3 DBF     1.18  Humid  <NA>             Ameriflux
##  3 US-DK2       36.0     -79.1 DBF     0.992 Humid  GENTLE SLOPE (<~ Ameriflux
##  4 US-DK3       36.0     -79.1 ENF     0.994 Humid  GENTLE SLOPE (<~ Ameriflux
##  5 US-IB2       41.8     -88.2 GRA     0.941 Humid  <NA>             Ameriflux
##  6 US-JO1       32.6    -107.  SHR     0.146 Arid   MEDIUM SLOPE (>~ Fluxnet
##  7 US-JO2       32.6    -107.  SHR     0.154 Arid   MEDIUM SLOPE (>~ Ameriflux
##  8 US-KM2       42.4     -85.3 GRA     0.950 Humid  UNDULATED/VARIA~ Ameriflux
##  9 US-KM3       42.4     -85.3 GRA     0.949 Humid  MEDIUM SLOPE (>~ Ameriflux
## 10 US-KON       39.1     -96.6 GRA     0.664 Humid  UNDULATED/VARIA~ Fluxnet
## # i 30 more rows
```

The Tables1 Section

This runs ANOVA, t-test, and description statistics, generate the information shown within the results, captions, and for tables 2, 3, 4, and S2:

Table 2

Descriptive statistics for DDF (hours) during drought and ND periods. Welch's t-test: t(17109.79) = -9.02, p < 0.001; [mean] = 0.21 h, ~13 min

filename: desc_Drought_DDF.csv; ttest_DDF.csv

Table 3

Descriptive statistics for DDF (hours) for each drought severity category (ND, LS, MS)

filename: desc_drought_cat_DDF.csv

Table 4

Descriptive statistics for DDF (hours) for each aridity category comparing each drought category (e.g., ND, drought)

filename: desc_AI_cat_DDF.csv

Table S2.

Summary statistical results for each vegetation (Veg) category

filename: desc_Veg_DDF.csv

A tibble: 2 x 10

Category group_var test t df p sig baseline comparison
1 Drought (Dr~ Drought t-te~ -9.02 17110. 2.10e- 19 *** Drought ND
2 Aridity cla~ AI_cat t-te~ -35.0 46984. 7.02e-265 *** Arid Humid
i 1 more variable: med_diff # A tibble: 2 x 9 Category group_var test F df1 df2 p sig omega_sq 1 Drought severity drought_cat ANOVA 117. 2 49641 2.11e-51 *** 0.00465 2 Vegetation type Veg ANOVA 575. 3 49640 0 *** 0.0335

Table 1: Descriptive stats by Drought (response = DDF). CI = 95% t-based CI of the mean.

Category	Group	sites_n	n_obs	mean	sd	median	ci_low	ci_high
Drought (Drought vs ND)	Drought	40	10957	-3.287	2.183	-3.5	-3.327	-3.246
Drought (Drought vs ND)	ND	40	38687	-3.075	2.099	-3.0	-3.096	-3.054

Table 2: Descriptive stats by drought_cat (response = DDF). CI = 95% t-based CI of the mean.

Category	Group	sites_n	n_obs	mean	sd	median	ci_low	ci_high	tukey
Drought severity	LS	40	8954	-3.403	2.212	-3.5	-3.449	-3.357	c
Drought severity	MS	15	2003	-2.766	1.966	-3.0	-2.852	-2.680	a
Drought severity	ND	40	38687	-3.075	2.099	-3.0	-3.096	-3.054	b

Table 3: Descriptive stats by Veg (response = DDF). CI = 95% t-based CI of the mean.

Category	Group	sites_n	n_obs	mean	sd	median	ci_low	ci_high	tukey
Vegetation type	DBF	7	8423	-2.604	1.950	-2.5	-2.646	-2.563	c
Vegetation type	ENF	8	7812	-3.010	2.150	-3.0	-3.057	-2.962	a
Vegetation type	GRA	14	17792	-2.942	2.004	-3.0	-2.972	-2.913	a
Vegetation type	SHR	11	15617	-3.662	2.204	-3.5	-3.697	-3.627	b

Table 4: Descriptive stats by AI_cat (response = DDF). CI = 95% t-based CI of the mean.

Category	Group	sites_n	n_obs	mean	sd	median	ci_low	ci_high
Aridity class	Arid	13	23228	-3.475	2.213	-3.5	-3.503	-3.447
Aridity class	Humid	27	26416	-2.811	1.982	-3.0	-2.835	-2.788

Table 5: t-tests

Category	group_var	test	t	df	p	sig	baseline	comparison	med_diff
Drought (Drought vs ND)	Drought	t-test	-9.0187	17109.79	0	***	Drought	ND	0.5
Aridity class	AI_cat	t-test	-34.9946	46984.06	0	***	Arid	Humid	0.5

Table 6: ANOVAs + omega squared

Category	group_var	test	F	df1	df2	p	sig	omega_sq
Drought severity	drought_cat	ANOVA	116.9620	2	49641	0	***	0.0047

Category	group_var	test	F	df1	df2	p	sig	omega_sq
Vegetation type	Veg	ANOVA	574.7901	3	49640	0	***	0.0335

The Tables2 Section

Drought severity - this runs ANOVA, t-test, and description statistics, generate the information shown within the results, captions, and for tables 4, 5, S3, and S4:

Table 4.

Descriptive statistics for DDF (hours) for each aridity category comparing each drought category (e.g., ND, drought)

filename: desc_AI_cat_x_drought_DDF.csv

Table 5.

Descriptive statistics for DDF (hours) for each aridity category comparing each drought severity category

filename: desc_AI_cat_x_drought_cat_DDF.csv

Table S3.

Summary statistic results for each vegetation (Veg) type under non-drought (ND) and drought conditions

filename: desc_Veg_x_drought_DDF.csv

Table S4.

Summary statistic results for each vegetation (Veg) type under non-drought (ND), less severe (LS) drought, and more severe (MS) drought conditions

filename: desc_Veg_x_drought_cat_DDF.csv

Table 7: Veg × drought (response=DDF); CI = 95% t-based CI of mean

Category	Veg	Drought	sites_n	n_obs	mean	sd	median	ci_low	ci_high
Veg	DBF	ND	7	6366	-2.533	1.947	-2.5	-2.581	-2.485
Veg	DBF	Drought	7	2057	-2.825	1.942	-3.0	-2.909	-2.741
Veg	ENF	ND	8	6123	-2.939	2.152	-3.0	-2.993	-2.885
Veg	ENF	Drought	8	1689	-3.268	2.124	-3.5	-3.369	-3.166
Veg	GRA	ND	14	13772	-2.920	1.977	-3.0	-2.953	-2.887
Veg	GRA	Drought	14	4020	-3.018	2.095	-3.0	-3.082	-2.953
Veg	SHR	ND	11	12426	-3.592	2.169	-3.5	-3.631	-3.554
Veg	SHR	Drought	11	3191	-3.933	2.317	-4.0	-4.013	-3.852

Table 8: t-tests (within each Veg): ND vs Drought (no p adjustment)

Category	Veg	t	df	p	p_adj	sig	baseline	comparison	med_diff
Veg	DBF	5.9333	3489.361	0.0000	0.0000	***	ND	Drought	-0.5
Veg	ENF	5.6184	2719.256	0.0000	0.0000	***	ND	Drought	-0.5
Veg	GRA	2.6255	6254.608	0.0087	0.0087	**	ND	Drought	0.0
Veg	SHR	7.4970	4725.072	0.0000	0.0000	***	ND	Drought	-0.5

Table 9: Veg × drought_cat (response=DDF); CI = 95% t-based CI of mean

Category	Veg	drought_cat	sites_n	n_obs	mean	sd	median	ci_low	ci_high
Veg	DBF	LS	7	1137	-2.834	1.879	-3.0	-2.943	-2.724
Veg	DBF	MS	6	920	-2.815	2.019	-3.0	-2.945	-2.684
Veg	DBF	ND	7	6366	-2.533	1.947	-2.5	-2.581	-2.485
Veg	ENF	LS	8	1487	-3.310	2.097	-3.5	-3.417	-3.204
Veg	ENF	MS	2	202	-2.953	2.295	-3.0	-3.271	-2.635
Veg	ENF	ND	8	6123	-2.939	2.152	-3.0	-2.993	-2.885
Veg	GRA	LS	14	3280	-3.137	2.162	-3.0	-3.211	-3.063
Veg	GRA	MS	4	740	-2.487	1.674	-2.5	-2.608	-2.366
Veg	GRA	ND	14	13772	-2.920	1.977	-3.0	-2.953	-2.887
Veg	SHR	LS	11	3050	-3.946	2.321	-4.0	-4.029	-3.864
Veg	SHR	MS	3	141	-3.642	2.224	-4.0	-4.012	-3.271
Veg	SHR	ND	11	12426	-3.592	2.169	-3.5	-3.631	-3.554

Table 10: ANOVAs (within each Veg): drought_cat (no p adjustment)

Category	Veg	F	df1	df2	p	sig	omega_sq
Veg	DBF	17.5792	2	8420	0	***	0.0039
Veg	ENF	18.0245	2	7809	0	***	0.0043
Veg	GRA	35.5763	2	17789	0	***	0.0039
Veg	SHR	31.6844	2	15614	0	***	0.0039

Table 11: Pairwise t-tests within each Veg: drought_cat vs ND (no p adjustment)

Category	Veg	drought_cat	t	df	p	p_adj	sig	baseline	med_diff
Veg	DBF	LS	-4.9475	1602.9633	0.0000	0.0000	***	ND	-0.5
Veg	DBF	MS	-3.9769	1179.7094	0.0001	0.0001	***	ND	-0.5
Veg	ENF	LS	-6.0990	2306.7793	0.0000	0.0000	***	ND	-0.5
Veg	ENF	MS	-0.0873	212.8303	0.9305	0.9305	ns	ND	0.0
Veg	GRA	LS	-5.2521	4670.8438	0.0000	0.0000	***	ND	0.0
Veg	GRA	MS	6.7896	853.6523	0.0000	0.0000	***	ND	0.5
Veg	SHR	LS	-7.6403	4445.8061	0.0000	0.0000	***	ND	-0.5
Veg	SHR	MS	-0.2624	143.0361	0.7934	0.7934	ns	ND	-0.5

Table 12: Pairwise t-tests within each AI_cat: drought_cat vs ND (no p adjustment)

Category	AI_cat	drought_cat	t	df	p	p_adj	sig	baseline	med_diff
AI_cat	Arid	LS	-7.0708	6448.4237	0.0000	0.0000	***	ND	-0.5
AI_cat	Arid	MS	-2.0577	39.1336	0.0463	0.0463	*	ND	-1.0
AI_cat	Humid	LS	-10.7944	6505.7730	0.0000	0.0000	***	ND	-0.5
AI_cat	Humid	MS	0.3382	2375.2236	0.7353	0.7353	ns	ND	-0.5

Table 13: Tukey HSD pairwise comparisons within each Veg: drought_cat

Category	Veg	comparison	group1	group2	diff	lwr	upr	p_adj	sig
Veg	DBF	LS-ND	LS	ND	-0.3010	-0.4479	-0.1541	0.0000	***
Veg	DBF	MS-ND	MS	ND	-0.2819	-0.4428	-0.1210	0.0001	***
Veg	DBF	MS-LS	MS	LS	0.0191	-0.1832	0.2214	0.9734	ns
Veg	ENF	LS-ND	LS	ND	-0.3717	-0.5171	-0.2263	0.0000	***
Veg	ENF	MS-ND	MS	ND	-0.0143	-0.3740	0.3454	0.9952	ns
Veg	ENF	MS-LS	MS	LS	0.3574	-0.0198	0.7345	0.0677	ns
Veg	GRA	LS-ND	LS	ND	-0.2171	-0.3082	-0.1260	0.0000	***
Veg	GRA	MS-ND	MS	ND	0.4331	0.2562	0.6101	0.0000	***
Veg	GRA	MS-LS	MS	LS	0.6502	0.4594	0.8410	0.0000	***
Veg	SHR	LS-ND	LS	ND	-0.3538	-0.4580	-0.2496	0.0000	***
Veg	SHR	MS-ND	MS	ND	-0.0494	-0.4861	0.3872	0.9620	ns
Veg	SHR	MS-LS	MS	LS	0.3044	-0.1397	0.7485	0.2429	ns

Table 14: Tukey HSD pairwise comparisons within each AI_cat: drought_cat

Category	AI_cat	comparison	group1	group2	diff	lwr	upr	p_adj	sig
AI_cat	Arid	LS-ND	LS	ND	-0.2727	-0.3589	-0.1866	0.0000	***
AI_cat	Arid	MS-ND	MS	ND	-0.7915	-1.6117	0.0286	0.0613	ns
AI_cat	Arid	MS-LS	MS	LS	-0.5188	-1.3417	0.3041	0.3017	ns
AI_cat	Humid	LS-ND	LS	ND	-0.3589	-0.4356	-0.2822	0.0000	***
AI_cat	Humid	MS-ND	MS	ND	0.0156	-0.0940	0.1252	0.9407	ns
AI_cat	Humid	MS-LS	MS	LS	0.3745	0.2490	0.5000	0.0000	***

Table 15: AI_cat × drought (response=DDF); CI = 95% t-based CI of mean

Category	AI_cat	Drought	sites_n	n_obs	mean	sd	median	ci_low	ci_high
AI_cat	Arid	ND	13	18699	-3.421	2.174	-3.5	-3.452	-3.390
AI_cat	Arid	Drought	13	4529	-3.698	2.355	-4.0	-3.767	-3.630
AI_cat	Humid	ND	27	19988	-2.752	1.971	-2.5	-2.779	-2.725
AI_cat	Humid	Drought	27	6428	-2.996	2.003	-3.0	-3.045	-2.948

Table 16: t-tests (within each AI_cat): ND vs Drought (no p adjustment)

Category	AI_cat	t	df	p	sig	baseline	comparison	med_diff
AI_cat	Arid	7.2138	6522.242	0	***	ND	Drought	-0.5
AI_cat	Humid	8.5486	10720.757	0	***	ND	Drought	-0.5

Table 17: AI_cat × drought_cat (response=DDF); CI = 95% t-based CI of mean

Category	AI_cat	drought_cat	sites_n	n_obs	mean	sd	median	ci_low	ci_high
AI_cat	Arid	ND	13	18699	-3.421	2.174	-3.5	-3.452	-3.390
AI_cat	Arid	LS	13	4489	-3.694	2.355	-4.0	-3.763	-3.625
AI_cat	Arid	MS	1	40	-4.213	2.431	-4.5	-4.990	-3.435
AI_cat	Humid	ND	27	19988	-2.752	1.971	-2.5	-2.779	-2.725
AI_cat	Humid	LS	27	4465	-3.111	2.017	-3.0	-3.170	-3.052
AI_cat	Humid	MS	14	1963	-2.736	1.945	-3.0	-2.822	-2.650

Table 18: ANOVAs (within each AI_cat): drought_cat (no p adjustment)

Category	AI_cat	F	df1	df2	p	sig	omega_sq
AI_cat	Arid	29.7805	2	23225	0	***	0.0025
AI_cat	Humid	61.6672	2	26413	0	***	0.0046